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Are potential bisphenol-A substitutes really safe for aquatic life? Impact on primary producers

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Öz

Bisphenol A threat to environmental health and human health and has been added to the Candidate List as Very High Concern Substances by the European Chemicals Agency. This led to the replacement of bisphenol A (BPA) with bisphenol analogues, which were considered "safer". However, there are very few scientific studies on the impact of BPA analogues on the environment. In this study, three analogues bisphenol B (BPB), bisphenol A diglycidyl ether (BADGE) and bisphenol F diglycidyl ether (BFDGE) were selected to investigate their ecotoxicological effects on the marine phytoplankton species Phaeodactylum tricornutum, which is representative of primary producers. Phaeodactylum tricornutum was exposed to different concentrations (0.5, 0.8, 1.0, 1.5, 2.0 mg/L) of BPB, BADGE and BFDGE analogues for 72 hours and the toxicity values of three BPA analogues were calculated by OECD 201 algal growth inhibition assay (IC50/EC50). In the light of the data obtained, algal growth inhibition (IC50/EC50) values for marine phytoplankton Phaeodactylum tricornutum were determined as 3.91 mg-BPA/L, 7.83 mg-BPB/L, 5.69 mg-BFDGE/L, 11.71 mg-BADGE/L. The results revealed that BPB, BFDGE and BADGE showed lower toxicity to Phaeodactylum tricornutum compared to BPA algal growth inhibition (3.91 mg-BPA/L). Therefore, it is necessary to share the results of the adverse effects of BPA analogues on aquatic organisms and to conduct ecotoxicological risk assessments.

Anahtar Kelimeler

Bisphenol analogues , marine phytoplankton , algal growth inhibition test , toxicity , Phaeodactylum tricornutum

Etik Beyan

No specific ethical approval was necessary for the study.

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Tüm Kaynakçayı Göster

Toplam **45** adet kaynakça vardır.

Ayrıntılar

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